```
A = [1,2,3;0,4,1;-1,5,-2]
```

```
A = 3×3
     1     2     3
     0     4     1
    -1     5    -2
```

```
norm(A,1)
```

```
ans =
    11
```

```
norm(A,inf)
```

```
ans =
     8
```

```
norm(A,2)
```

```
ans =
   6.724002372068454
```

```
norm(A,'fro')
```

```
ans =
   7.810249675906654
```

```
sum(sum(abs(A).^2))^(1/2)
```

```
ans =
   7.810249675906654
```

Sistemul

Ax = b

A, b date de intrare, x de iesire

x = f(A,b) = $A^{-1}b$

$f : \Re^{nxn} x \Re^{nx1} \rightarrow \Re^{nx1}$

$\text{cond}_f((A,b)) = ||A|| * |A^{-1}||$

Problema 1

$$\begin{bmatrix} 10 & 7 & 8 & 7 \\ 7 & 5 & 6 & 5 \\ 8 & 6 & 10 & 9 \\ 7 & 5 & 9 & 10 \end{bmatrix} * x = \begin{bmatrix} 32 \\ 23 \\ 33 \\ 31 \end{bmatrix}$$

```
A=[10, 7, 8, 7;7,5,6,5;8,6,10,9;7,5,9,10];
b=[32;23;33;31];
```

```
xe = A\b
```

```
xe = 4×1
     1
     1
     1
     1
```

Perturbăm membrul drept $b_p = \begin{bmatrix} 32.1 \\ 22.9 \\ 33.1 \\ 30.9 \end{bmatrix}$

```
bp = [32.1;22.9;33.1;30.9];
xp1=A\bp
```

```
xp1 = 4×1
   9.199999999999680
 -12.599999999999463
   4.499999999999858
  -1.099999999999915
```

```
eri1 = norm(b-bp)/norm(b);
ero1=norm(xe-xp1)/norm(xe);
factor_aplificare = ero1/eri1
```

```
factor_aplificare =
     2.460567236431338e+03
```

Perturbam matricea $A_p = \begin{bmatrix} 10 & 7 & 8.1 & 7.2 \\ 7.08 & 5.04 & 6 & 5 \\ 8 & 5.98 & 9.89 & 9 \\ 6.99 & 4.99 & 9 & 9.98 \end{bmatrix} * x = \begin{bmatrix} 32 \\ 23 \\ 33 \\ 31 \end{bmatrix}$

```
Ap = [10, 7, 8.1, 7.2 ; 7.08,5.04,6,5; 8,5.98,9.98, 9;6.99,4.99,9,9.98];
xp2 = Ap\b
```

```
xp2 = 4×1
  -5.792243651257218
  12.018809315278880
  -1.565554294241548
   2.565522373468965
```

```
eri2 = norm(A-Ap)/norm(A);
ero2=norm(xe-xp2)/norm(xe);
factor_aplificare = ero2/eri2
```

```
factor_aplificare =
     8.963806327142571e+02
```

Raportul mare dintre eroarea relativa la iesire si eroarea relativa la intrare se aplica prin numarul de conditionare $\text{cond}_A = ||A|| * ||A^{-1}||$

```
cond(A)
```

```
ans =
     2.984092701675700e+03
```

```
det(A)
```

```
ans =
   1.000000000000049
```

```
inv(A)
```

```
ans = 4×4
  25.000000000000576 -41.000000000000938  10.000000000000227  -6.000000000000136
 -41.000000000000938  68.000000000001535 -17.000000000000369  10.000000000000222
  10.000000000000227 -17.000000000000369   5.000000000000089  -3.000000000000053
  -6.000000000000136  10.000000000000222  -3.000000000000053   2.000000000000032
```

Problema 2

Matricea Hilbert

$$H_{ij} = \frac{1}{i+j+1}$$

H_4 este

```
sym(hilb(4))
```

```
ans =
```

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \end{pmatrix}$$

```
sym(invhilb(4))
```

```
ans =
```

$$\begin{pmatrix} 16 & -120 & 240 & -140 \\ -120 & 1200 & -2700 & 1680 \\ 240 & -2700 & 6480 & -4200 \\ -140 & 1680 & -4200 & 2800 \end{pmatrix}$$

$$\mathrm{cond}_2(H_n) = \lambda_{\max}(H_n)\lambda_{\max}\left(H_n^{-1}\right)$$

O estimare teoretica pentru n mare este : $\mathrm{cond}_2(H_n) = \dfrac{\left(1+\sqrt{2}\right)^{4n+4}}{2^{\frac{15}{4}}\sqrt{\pi n}}$

```
for n=10:10:100
```

```
    c = max(eig(hilb(n)))*max(eig(invhilb(n)));
    th = (sqrt(2)+1)^(4*n+4)/(2^(15/4)*sqrt(pi*n));
    fprintf("%3d %11.4e %11.4e\n",n,c,th)
end
```

```
 10  1.6026e+13  9.2192e+14
 20  2.4522e+28  1.3342e+30
 30  4.2278e+43  2.2294e+45
 40  7.6529e+58  3.9513e+60
 50  1.4229e+74  7.2330e+75
 60  2.6913e+89  1.3513e+91
 70 5.1508e+104 2.5604e+106
 80 9.9444e+119 4.9016e+121
 90 1.9328e+135 9.4578e+136
100 3.7765e+150 1.8363e+152
```

```
H=hilb(10);
b=H*ones(10,1);
format long
x=H\b
```

```
x = 10×1
   0.999999999095413
   1.000000077638004
   0.999998354294758
   1.000014906295676
   0.999929106136442
   1.000194425970870
   0.999681635015205
   1.000307148615796
   0.999838981464458
   1.000035366078449
```

```
warning off
for n=[10,20,40,80]
    t = linspace(-1,1,n);
    V=vander(t);
    et=1/pi*exp(-pi/4)*exp(n*(pi/4+1/2*log(2)));
    x=[n,norm(V,inf)*norm(inv(V),inf),condest(V),et];
    fprintf("%3d %e %e    %e\n",x)
end
```

```
 10 2.056171e+04 1.362524e+04    1.196319e+04
 20 1.751063e+09 1.053490e+09    9.861382e+08
 40 1.208386e+19 6.926936e+18    6.700689e+18
 80 3.045101e+37 2.392380e+37    3.093734e+38
```

```
warning on

warning off
for n=10:15
    t=1./(1:n);
    V=vander(t);
    x=[n, norm(V,inf)*norm(inv(V),inf), condest(V), n^(n+1)];
    fprintf("%3d %e %e    %e\n", x)
end
```

```
 10 5.792417e+11 5.905580e+11    1.000000e+11
 11 2.382382e+13 2.278265e+13    3.138428e+12
 12 1.060780e+15 9.692982e+14    1.069932e+14
 13 5.087470e+16 4.732000e+16    3.937376e+15
 14 2.615990e+18 2.419006e+18    1.555681e+17
 15 1.436205e+20 1.294189e+20    6.568408e+18
```

```
warning on
```

$P = a_0x^n + a_1x^{n-1} + \ldots + a_n$

P=? a.i. $P(x_i) = b_i$

<= Vandermonde

Problema 4

```
n=20;
v=1:20;
P=poly(v);
roots(P)
```

```
ans = 20×1
  19.999874055724192
  19.001295393676987
  17.993671562737585
  17.018541647321989
  15.959717574548915
  15.059326234074415
  13.930186454760916
  13.062663652011070
  11.958873995343460
  11.022464271003383
      ⋮
```

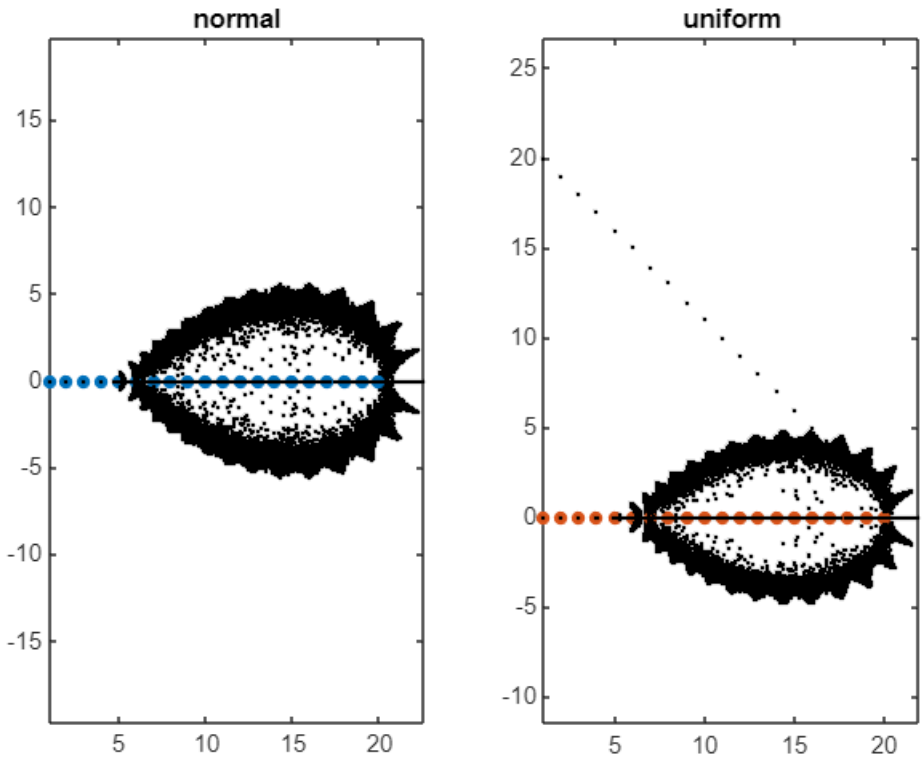
```
function nc=condpol(p,xi)
    if nargin < 2
        xi = roots(p);
    end
    n=length(p)-1;
    dp=(n:-1:1).*p(1:end-1);
    nc=1./(abs(xi.*polyval(dp,xi))).*(polyval(abs(p(2:end)),abs(xi)));
end

function wilkinson(n)
    p=poly(1:n);
    h=plot(1:n,zeros(1,n),'.');
    set(h,'Markersize',15);
    hold on
    for k=1:1000
        r = randn(1,n+1);
        pr=p.*(1+1e-10*r);
        z=roots(pr);
        h2=plot(z,'k.');
```

```
        set(h2,'Markersize',4)
    end
    axis equal
end

function wilkinsonu(n)
    p=poly(1:n);
    h=plot(1:n,zeros(1,n),'.');
    set(h,'Markersize',15);
    hold on
    for k=1:1000
        r = (1e-10)*rand(1,n+1);
        pr=p + r.*p;
        z=roots(pr);
        h2=plot(z,'k.');
        set(h2,'Markersize',4)
    end
    axis equal
end

subplot(1,2,1)
wilkinson(20)
title('normal')
subplot(1,2,2)
wilkinsonu(20)
title('uniform')
```



Problema 5

```
function E=recdir(n)
    E=1/exp(1);
    for k=2:n
        E = 1-k*E;
    end
end

for n=2:25
    E = recdir(n);
    fprintf("E(%2d)=%+6.4e\n",n,E)
end
```

```
E( 2)=+2.6424e-01
E( 3)=+2.0728e-01
E( 4)=+1.7089e-01
E( 5)=+1.4553e-01
E( 6)=+1.2680e-01
E( 7)=+1.1238e-01
E( 8)=+1.0093e-01
E( 9)=+9.1612e-02
E(10)=+8.3877e-02
E(11)=+7.7352e-02
E(12)=+7.1773e-02
E(13)=+6.6948e-02
E(14)=+6.2731e-02
E(15)=+5.9034e-02
E(16)=+5.5459e-02
E(17)=+5.7192e-02
E(18)=-2.9454e-02
E(19)=+1.5596e+00
E(20)=-3.0192e+01
E(21)=+6.3504e+02
E(22)=-1.3970e+04
E(23)=+3.2131e+05
E(24)=-7.7114e+06
E(25)=+1.9279e+08
```

$E_n = 1 - nE_{n-1} = 1 - n(1 - (n-1)E_{n-2}) = 1 - n + n(n-1)E_{n-2} = \ldots = P(n) + (-1)^n n! E_1$

$$\text{cond}(f)(E_1) = \frac{|E_1||\frac{\partial f}{\partial E_1}|}{|E_n|} > \frac{E_1}{E_1} n! = n!$$

Se aplica in ordinea inversa $E_{n-1} = \frac{1 - E_n}{n}$

$$E_n = \frac{1 - E_{n+1}}{n+1} = \frac{1 - \frac{1 - E_{n+2}}{n+2}}{n+1} = \frac{1}{n+2} + \frac{E_{n+2}}{(n+1)(n+2)} = \frac{n+2}{(n+3)(n+1)} - \frac{E_{n+3}}{(n+1)(n+2)(n+3)} = Q(n) + (-1)^k \frac{E_{n+k}}{(n+1)\ldots(n+k}$$

$$\text{cond}(g)(E_{n+k}) = \frac{\left|E_{n+k}\frac{\partial g(E_{n+1})}{\partial E_{n+k}}\right|}{|E_n|} = \frac{E_{n+k}}{E_n}\frac{1}{(n+1)\ldots(n+k)} < \frac{E_{n+k}}{E_{n+k}}\frac{1}{(n+1)\ldots(n+k)} = \frac{1}{(n+1)\ldots(n+k)}$$

cu cat k e mai mare, cu atat factorul este mai mic

```
function E=recinv(n,k)
    E =0;
    for j =n+k:-1:n+1
        E=(1-E)/j;
    end
end

for k = 10:18
    E=recinv(1,k);
    err=abs(E-exp(-1))/exp(-1);
    fprintf("k=%2d, err=%e\n",k, err);
end
```

```
k=10, err=5.267586e-09
k=11, err=4.073053e-10
k=12, err=2.922473e-11
k=13, err=1.955900e-12
k=14, err=1.228285e-13
k=15, err=7.242958e-15
k=16, err=4.526849e-16
k=17, err=0.000000e+00
k=18, err=0.000000e+00
```